Asdell's Patterns of Mammalian Reproduction

A Compendium of Species-Specific Data

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We dedicate this book to Kenneth E. Wing, without whose administrative support it would not have survived beyond an early embryonic stage. We also dedicate this book to our teachers, who represent the past; our colleagues, who represent the present; and our students, who represent the future. Their inspiration, knowledge, and labor provided the foundation on which this book is based.

VH adds a personal dedication to the memory of her brother, Sandy.

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Preface

More than fifty years ago Sydney A. Asdell, one of the great minds of the early years of reproductive physiology, undertook the difficult task of compiling the basic facts of reproductive anatomy and physiology for all mammals. The results appeared in his much praised and frequently consulted *Patterns of Mammalian Reproduction*, published in 1946. The discerning intellect of this man is apparent in the introductory section to the book, a section he titled "The Scaffolding".

For the second edition, published in 1964, Dr. Asdell changed the book from an introduction to mammalian reproductive physiology to an encyclopedia of reproductive data. The introductory scaffolding was razed to clear the way for a more extensive compilation of species-specific data on mammals. Interpretive accounts detailing physiological mechanisms were still included for a diverse array of mammals, but the overall goal was to provide the basic information available on the reproduction of individual species.

In 1976 Dr. Asdell asked Ari van Tienhoven to prepare a revision of the book. Dr. van Tienhoven initially declined, intending to find a replacement, then accepted in 1984 when 1) a sabbatical leave provided the opportunity to initiate the collection of data, 2) his efforts to find someone to revise the book failed, and 3) his two coauthors — Virginia Hayssen and Ans van Tienhoven — as well as Kenneth E. Wing, Associate Dean of the College of Agriculture and Life Sciences at Cornell University, assured him that the task of collecting the data was not insurmountable. When it proved to be more formidable than originally imagined, Dr. van Tienhoven had due cause to doubt the veracity of his coauthors. But with the book's final phases under way eight years later, their encouragement seems justified.

As the scientific literature has expanded, the scope of each edition has contracted. Empirical analyses have replaced physiological interpretations, and the emphasis is on a neutral presentation of the published facts of mammalian reproduction. The scope of this revision is similar in two main ways. First, we do not provide extensive interpretations of physiological mechanisms for each taxon as Dr. Asdell did. Second, we have not included data on domestic and laboratory species, as a voluminous literature is readily available. For instance, J. Vaissaire's review of the reproduction of laboratory and domestic animals summarizes some 6000 references (Sexualité et reproduction des mammifères domestiques et de laboratoire [Paris: Maloine, 1987]).

To compile this revision we examined over 150 core journals (see Appendix 2), gleaning data as well as referrals to other sources. We chose articles whose titles and/or abstracts suggested the presence of data on reproductive parameters. In all, over 20,000 sources were reviewed. We used primary sources except when they were not accessible or when we could not translate the language in which they were published. For Russian, Finnish, Chinese, and Japanese publications, we read English abstracts, summaries, or translated secondary sources. Even though some review articles (e.g., field guides, regional accounts) do not cite sources, we included data from such articles if these data were the sole information we could find or if the information differed from that in the primary sources we cite.

Data on a selected set of physical, ecological, and reproductive parameters were culled from each article. These data were coded for computer entry. After extensive checking for errors, we sorted the files and reformatted the appropriate data into tables or summarized them for the ordinal and familial accounts. Bibliographic information for each reference was also computer coded and entered into bibliographic data files. Although we proofread the manuscript at every stage of its compilation and endeavored to be thorough in our survey of the field, errors and omissions certainly exist. We would appreciate notification of any errors you find, so that they may be corrected in future printings.

This book has been a labor of love not only for us but also for the many who assisted us. We thank Kenneth E. Wing, Associate Dean of the College of Agriculture and Life Sciences of Cornell University[1], for his important support in the initial stages of our undertaking and for his sustained interest.

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[1] As of 1 December, 1992, President of the State University of New York at Cobleskill, NY.

The collection of the data would have been less thorough if it were not for the crucial support we received from a multitude of librarians. The reference librarians at Mann Library, Cornell University, and the staff librarians at the Smith College Science Library were always willing to go the extra mile to find articles for which we had incomplete citations or to find information on obscure species. The collegiality of these librarians has been an example of professionalism and a source of friendship. The librarians handling interlibrary loans at both Cornell University and Smith College were equally helpful and patient with us, as were the desk librarians, who were always ready to search for publications which were not in their appointed places. J. Craig of the Morrill Science Library, University of Massachusetts at Amherst, spent many, many hours of his time finding and copying references. His generosity and dedication were boundless and are admired as well as appreciated. We are also grateful to J.P. Hearn and the librarians of the Zoological Society of London for making the van Tienhovens' sabbatical stay in the society's library a pleasant and profitable one.

We thank the librarian of the Library of the Royal Society of London for his cordial welcome and the free use of the collection. Use of the research library at the Brookfield Zoo, Brookfield, Ill., as well as much appreciated desk space, was made possible by G. Rabb, M. Rabb, P. Parker, and R. Lacy. A. Petric of the Brookfield Zoo supplied gestation lengths for some hoofed stock from the zoo's records. M. Jones and K. Benirschke provided access to data from the San Diego Zoological Society's records as well as helpful advice. L. Gordon smoothed our use of the mammal collection at the National Museum of Natural History. We are indebted to C. and E. James, Adelaide, Australia, J. Wright, Carlisle, Penn., and the late W. Wimsatt, Ithaca, N.Y., for access to their personal libraries.

We are grateful to the following colleagues who graciously sent us compilations of their reprints for review or provided data from obscure sources: K. Armitage, R.T.F. Bernard, K. Bhatnagar, H.E. Broadbooks, G.N. Cameron, L.N. Carraway, J.A. Chapman, D. Dewsbury, J.E.C. Flux, M. Gosling, P.D. Heideman, N.R. Holler, D.D. Hopkins, B.E. Horner, H. Ikeda, J.U.M. Jarvis, U. Judes, J. Kenagy, M.L. Kennedy, D. Klingener, J.A. Lackey, J. Murie, D.K. Odell, R.H. Pine, A.I. Roest, L. Selwood, M. Tuttle, T.A. Uchida, T. Vaughan, B.J. Verts, and D.W. Walton.

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The scientific names of mammals change over time. We are in debt to M. Carleton, National Museum of Natural History, Smithsonian Institution, K.A.R. Kennedy, Section of Ecology and Systematics, Cornell University, K. Koopman, American Museum of Natural History, and D.K. McClearn, Section of Ecology and Systematics, Cornell University, for their help in resolving many of our most intransigent taxonomic synonymies. Their professional courtesy and gracious willingness to decipher the systematic lexicon are a credit to their field.

The patience and help of T. Ferrara, Cornell University, and S. Beaumier, Smith College, in word processing and data entry are gratefully acknowledged. The many work-study students who helped code and enter bibliographic and scientific data also deserve our sincere thank-you.

The financial support of Cornell University, Smith College, and the Friends of the Smith College Library for clerical assistance, work-study students, library fees, and supplies are gratefully acknowledged.

VH would like to acknowledge the personal support and encouragement of her brother Sandy who had as many views of the bright side of life as there are numbers in this book.

Finally, we owe a debt to Cornell University Press and our editor, Robb Reavill, for the years of commitment to this book.

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Organization of the Book

In the almost thirty years since the appearance of the second edition of *Patterns of Mammalian Reproduction*, the field of reproductive physiology has expanded dramatically. Because we could not cover in detail all the developments and still thoroughly review the basics of reproduction across mammals, we have chosen the latter course. To compensate we have provided lists of citations arranged by topic to the literature on the endocrinology, reproductive anatomy, and reproductive physiology of each family. In addition, citations pertaining to domestic and common laboratory species have been curtailed.

As in previous editions, the data are organized taxonomically. The pattern of reproduction is summarized for each mammalian order and family. The familial accounts also include data on neonatal development and number of mammae, summarized by genus. Following each familial account is a tabular listing of species-specific data for neonatal mass and size, weaning mass and size, litter size, age at sexual maturity, estrous cycle length, gestation length, lactation length, number of litters per year and/or interbirth (interlitter) interval, and seasonality of reproduction. Each reproductive variable is listed separately. Genera, and species within each genus, are arranged alphabetically. Where appropriate, data are ordered by sequence of the reproductive cycle; for example, the listings for sexual maturity provide estimates of the age at first conception before those for the age at first birth. Data on length, size, and mass are ordered numerically from smallest to largest. Literature sources for each entry are provided. In sum, the published data for a given species are listed by reproductive variable for each family. This organization makes finding information for any one species more difficult but allows easier comparisons among species as well as easier extrapolations to species for which no data exist.

What taxonomy was used?

Whenever possible, our taxonomy follows *Mammal Species of the World* by J.H. Honacki, K.E. Kinman, and J.W. Koeppl. M. Carleton (National Museum of Natural History, Smithsonian Institution), K.A.R. Kennedy (Section of Ecology and Systematics, Cornell University), K. Koopman (American Museum of Natural History), and D.K. McClearn (Section of Ecology and Systematics, Cornell University) provided expert assistance in resolving taxonomic synonymies as did colleagues at the 1991 meeting of the American Society of Mammalogists. Notwithstanding these efforts, nomenclatural errors may still exist, especially for species described in older references. Questionable taxonomy is indicated by a question mark after the species name.

Medical journals often do not provide the scientific name of the study subjects. When such publications contained useful data, we made the following nomenclatural assignments: baboon (*Papio hamadryas*), fox (*Vulpes vulpes*), ferret (*Mustela putorious*), mouse (*Mus musculus*), rat (*Rattus norvegicus*), and rabbit (*Oryctolagus cuniculus*).

To include as much information on little-known species as possible, we generally do not list data for domesticated species — dog (*Canis familiaris*), cat (*Felis catus*), horse (*Equus caballus*), pig (*Sus scrofa*), cow (*Bos taurus*), sheep (*Ovis aries*), goat (*Capra hircus*) — and common laboratory animals — rhesus macaque (*Macaca mulatta*), lab mouse (*Mus domesticus*), white rat (*Rattus norvegicus*), Mongolian gerbil (*Meriones unguiculatus*), golden hamster (*Mesocricetus auratus*), Siberian hamster (*Phodopus sungorus*), guinea pig (*Cavia porcellus*), and white rabbit (*Oryctolagus cuniculus*). Data from their wild counterparts are included.

What if you do not know the family of the animal?

An index of mammalian species provides the page number for the appropriate family.

What if you do not know the scientific name of the animal?

An index of the common names of over 300 well-known mammals provides the page number for the appropriate family.

What data are in the familial summaries?
Each familial account includes the following information when available:

- Common name of family
- Alphabetic listing of all the genera in the family with an estimate of their adult mass or head-body length if mass data were not available. For a few genera, neither estimate of adult size was found.
- General habitat, diet, and geographical distribution of family
- Sexual dimorphism of mass (by genus if appropriate and available)
- Presence of specialized reproductive features
  - anatomical or functional asymmetry
  - induced ovulation
  - copulatory plug, copulatory tie
- Placental morphology (layers and shape)
- Implantation (location and depth)
- Uterine shape
- Presence of placentophagia or postpartum estrus
- Developmental state of neonate (by genus if appropriate)
  - furred or naked at birth
  - blind or time to eye opening
- Number of mammae (by genus if appropriate)
- Male reproductive anatomy
- Citations, by topic, of publications which detail reproductive structure and/or function

What do the citations grouped at the end of each familial account cover?
In reviewing the literature we read many papers which detailed the structure or function of reproduction in various species, but which did not have data appropriate for inclusion in the tables. We grouped citations to such papers by topic and listed them at the ends of the familial accounts. Thus these bibliographical listings are not exhaustive for a given topic, but instead cite the literature we encountered in our search for reproductive data. In general, we categorized papers into the following topics: **female anatomy** (histology, fine structure, or development of ovary, ovarian follicles, corpora lutea, uterus, ducts, reproductive glands including mammary glands); **male anatomy** (histology, fine structure, or development of testes, accessory glands, ducts, or secondary sex characteristics); **endocrinology** (reproductive hormone levels, pharmacological experiments, gonadectomy experiments, hypophysectomy experiments, binding proteins, uterine reactions, corpora lutea function); **spermatogenesis** (spermiogenesis, testicular blood flow, sperm morphology, seasonal changes in testicular morphology, physiology, or function); **copulation** (courtship, sexual behavior, effects of hormones on sexual behavior); **milk composition**; **reproductive energetics** (energetic cost of lactation or gestation, metabolic rates during reproduction); and **light** (effects of photoperiod, melatonin levels, and pinealectomy on reproductive processes). For some families other topics (or subtopics) are included.

What are the sources of the unannotated information?
Unannotated information came from the following sources or from reviews cited in them — common names: Corbet and Hill 1980, Nowak and Paradiso 1983, Nowak 1991; character synopses: Anderson and Jones 1984; natural history: *Grzimek's Encyclopedia of Mammals* 1990, Nowak and Paradiso 1983, Nowak 1991, accounts in Mammalian Species and regional compilations; sex chromosomes: Fredga 1970. The American Society of Mammalogists has synthesized the literature on over 400 mammalian species in separate short pamphlets in their series Mammalian Species. These are listed by taxon in Appendix 1.

What species-specific data tables are provided for each family?
Tables of neonatal mass, neonatal length, weaning mass, weaning length, age at sexual maturity, estrous cycle length, gestation length, lactation length, number of litters per year and/or interbirth/interlitter interval, and seasonality of reproduction are provided for each family if appropriate data were available.

How are the data organized within each table?

For each reproductive category, the data are organized alphabetically by genus, and alphabetically by species within each genus. For each species, independent estimates are listed separately and the citations given. All data are numerically ordered for each species from smallest to largest, but some are grouped into subcategories, as follows:

NEONATAL MASS, WEANING MASS

- Neonate mass refers to mass at birth
- Weaning-mass data are accompanied by an age when reported
- Sex is noted when appropriate

NEONATAL LENGTH, WEANING LENGTH

- Neonate length refers to length at birth
- Weaning-length data are accompanied by an age when reported
- Sex is noted when appropriate
- Crown-rump (CR) or head-body (HB) lengths are indicated if known (unfortunately, many published data do not indicate what was measured: CR, HB, or total length)

LITTER SIZE

- Units (e.g., embryos, corpora lutea, or young) provided when available

AGE AT SEXUAL MATURITY

- Female (f) data first, then male (m) data, then unknown or both (f/m)
- For each sex data are grouped by sequence of occurrence

- for females:	- for males:
1st estrus	large testes
1st mating	1st spermatogenesis
1st conception	1st sperm
1st pregnant	1st mating
1st birth	1st sire offspring

ESTROUS CYCLE LENGTH

- Portions of the cycle are listed first (i.e. follicular phase, luteal phase, menses, proestrus, estrus, metestrus, diestrus, anestrus) followed by length of entire cycle

GESTATION LENGTH

- Time of implantation first, followed by full gestation length with gestational delays noted

LACTATION LENGTH

- The following measures of lactation length are reported in an approximate sequence of occurrence (unfortunately, many sources did not state the measure used):
  - teat attachment
  - den emergence
  - 1st solid food
  - pouch exit
  - duration of nursing, suckling, lactation
  - weaning

NUMBER OF LITTERS PER YEAR or INTERBIRTH/INTERLITTER INTERVAL

- Interbirth/interlitter intervals with and without lactation are indicated if appropriate

SEASONALITY OF REPRODUCTION

- The months or time of year when specific reproductive stages occur are presented along with where the data were collected
- The following measures of seasonality are used:
  - testes large
  - mating season
  - pregnant females present
  - birth season
  - lactating females present
  - young found
  - reproductive period

- Ambiguous citations to "breeding seasons" were assumed to mean "reproduction occurs" and are listed as "repro" in the tables
- For each species, the above measures are listed by sequence of occurrence
- Within each measure, data are listed chronologically by month or time of year

How is each line of data organized?

For each line of data the mean (numerical average) is presented first, followed by the standard deviation (sd), the standard error (se), the median or mode, the range (r), and the sample size (n). Unspecified estimates of variability are indicated by ±. Confidence intervals are indicated by ci and reported verbatim from the references, i.e. if only one number rather than an interval was provided, only one number is reported here.

English measures were converted to metric and usually rounded in such a way as to retain the accuracy of the original data.

Data from the same reference may be given for more than one entry if they represent different measures, treatments, populations, or subspecies, and if the samples were large for all groups or the data appeared to be, or were, statistically different.

Identical data from different references may be merged to one entry if the data were from the same author, the data were from secondary sources, or the data were from different authors who used each other's work. Similar data from different references may be merged in one entry if they were from the same author and not significantly different. The largest sample size is reported if similar data were merged.

Some data were not used. Data published before World War I which conflicted with those published after World War II were deleted. Data from review articles were not used if they duplicated data from the primary literature. If many more accurate data were available, the following may have been deleted: anecdotal data, data with small sample size (n1-2), data given without methods, data given without estimate of variability, data given without sample size.

Some words of caution

The data are only as good as their sources, and the sources are quite variable. We accurately recorded what has been published, but what has been published may not be accurate. We spent eight years collecting and formatting the data. Resolving all the controversy within them could take another eight years, at least. If a particular value is critically important, please establish its veracity by direct observation.

The adult mass data may be for only the best known species or for only a few specimens and should be considered approximate. Most adult-mass data are from the publications we reviewed, augmented by general sources, such as *Walker's Mammals of the World*, *Grzimek's Encyclopedia of Mammals*, or regional accounts.

Finally, we would like to make a plea to all those readers who will write, edit, or review manuscripts which have data suitable for a compendium such as this. Please make sure that units are included with each measure and that an indication is clearly made as to which estimate of variation was used.

Abbreviations

ca	corpus albicans
cl	corpus luteum
cm	centimeter
CR	crown-rump length
d	day
ea	each
emb	embryo or fetus
est	estimate
f	female
FA	forearm length (Chiroptera)
g	gram
gest	gestating, gestation
h	hour
HB	head-body length
ibi	interbirth or interlitter interval
kg	kilogram
lact	lactating, lactation
lt	litter
m	male or meter
mm	millimeter
mo	month
ppe	postpartum estrus
preg	pregnant
ps	placental scars
repro	reproduction
rpy	removal of pouch young
TL	total length
vag perf	vaginal perforation
wk	week
yg	young
yr	year

statistics

~	approximately
<	less than
≤	equal to or less than
>	greater than
≥	equal to or greater than
±	unknown variance estimate
ci	confidence interval
est	estimate
min	minimum
max	maximum
n	sample size
r	range
sd	standard deviation
se	standard error

seasonality

fall	fall, autumn
repro	reproduction occurs
spr	spring
sum	summer
win	winter
yr rd	year round

months

Jan	January
Feb	February
Mar	March
Apr	April
May	May
June	June
July	July
Aug	August
Sept	September
Oct	October
Nov	November
Dec	December

location

Co	county
E, e	East (eastern)
I	island
N, n	North (northern)
R	river
S, s	South (southern)
W, w	West (western)

Canadian Provinces/Territories

ALB	Alberta
BC	British Columbia
MAN	Manitoba
NB	New Brunswick
NEW	Newfoundland
NS	Nova Scotia
NWT	Northwest Territories
ONT	Ontario
PEI	Prince Edward Island
QUE	Quebec
SAS	Saskatchewan
YUK	Yukon

Australian States/Territories

ACT	Australian Capital Territory
NSW	New South Wales
NT	Northern Territory
QLD	Queensland
SA	South Australia
TAS	Tasmania
VIC	Victoria
WA	Western Australia

Countries

UK	United Kingdom
US	United States of America
USSR	Union of Socialist Soviet Republics (used when we were uncertain as to which new republic was appropriate)

US States

AK	Alaska
AL	Alabama
AR	Arkansas
AZ	Arizona
CA	California
CO	Colorado
CT	Connecticut
DC	District of Columbia
DE	Delaware
FL	Florida
GA	Georgia
HI	Hawaii
IA	Iowa
ID	Idaho
IL	Illinois
IN	Indiana
KS	Kansas
KY	Kentucky
LA	Louisiana
MA	Massachusetts
MD	Maryland
ME	Maine
MI	Michigan
MN	Minnesota
MO	Missouri
MS	Mississippi
MT	Montana
NC	North Carolina
ND	North Dakota
NE	Nebraska
NH	New Hampshire
NJ	New Jersey
NM	New Mexico
NV	Nevada
NY	New York
OH	Ohio
OK	Oklahoma
OR	Oregon
PA	Pennsylvania
RI	Rhode Island
SC	South Carolina
SD	South Dakota
TN	Tennessee
TX	Texas
UT	Utah
VA	Virginia
VT	Vermont
WA	Washington
WI	Wisconsin
WV	West Virginia
WY	Wyoming

Lactation is the quintessence of mammals and the production of milk from mammary glands distinguishes mammals from all other animals. The three major taxa of mammals, Monotremata, Eutheria, and Marsupialia, are also characterized by their reproductive biology. The monotremes (Prototheria or Monotremata) are the only mammals which lay and incubate eggs, while both the Eutheria and the Marsupialia are viviparous. After very short gestation lengths, marsupials give birth to tiny offspring which depend on milk for most of their growth and development. Consequently, lactation in marsupials is unusually complex and its physiological control is highly evolved. In contrast, eutherians use gestation to provide a large portion of their maternal investment and the placental structures of eutherians are highly derived.

The length of lactation varies almost three orders of magnitude from 4 d (*Cystophora*, hooded seal) to over 900 d (Pongidae). In utero development (gestation or pregnancy) is less variable and ranges from 10-14 d in several marsupials to over 650 d in elephants. Most mammals give birth to between 1 and 15 young at intervals ranging from 3-4 wk for many rodents to 3-4+ years for sirenians, elephants, and rhinos among others. The highly specialized monotremes lay 1-3 eggs every 1-2 years.

Mammalian blastocysts are bilaminar with a trophoblast. Embryo formation is either from the outer layers of the blastocyst (monotremes and marsupials) or from the inner cell mass (eutherians). Embryonic Wolffian ducts differentiate into the male reproductive system, while Müllerian ducts form the female reproductive tract and are medial to the ureters in eutherians. Both sexes have paired gonads which secrete steroid hormones and produce gametes. The uterine lining (endometrium) and the placenta have endocrine as well as nutritive functions. Reproductive products may exit the body directly or in combination with renal metabolites via a urogenital sinus or with both renal and digestive material via a cloaca.

Females have 1 (probably 2, *Distoechurus*) to 29 (*Tenrec*) mammary glands as well as a pair of oviducts and uteri. Uteri may be fused along some (bipartite) or all (simplex) of their length. The utero-tubal junction separates the oviducts from the uteri, while the cervix separates the uterus from the terminal portion of the female tract (the vagina). Female monotremes and eutherians have a single vagina, whereas the structure is paired in the Marsupialia.

Male epididymides and deferent ducts are paired, but fuse before they pass through the penis which may be adorned with diverse protruberances, split along some or most of its length, or supported with a bone (os penis or baculum). The number and type of accessory glands associated with the male reproductive tract vary, although most males have at least a pair of bulbourethral glands and a single prostatic gland (only additional glands are noted in the familial accounts). Males are heterogametic structure. Sex chromosomes and sex differentiation vary (3492, 4013), although the female sex chromosome is highly conserved (10261a, 11474a, 11474b, 11474c). Male monotremes, eutherians, and didelphid marsupials have mammary glands (9045a, 9190a).

Several recent synthetic reviews of mammalian reproductive biology exist (464, 1317, 5891, 6165, 11190a). Comparative reviews exist for a few topics: implantation (505, 3010, 9461, 11803), sperm (710, 2169), environmental control (1316), fetal growth rates (3490), gestation (5726a), corpus luteum regulation (9312), birth (7775, 10804), litter-size variability (7698, 12062), origin of lactation (971), lactation (4523a, 5291, volumes 41 and 51 Symp Zool Soc London), and milk composition (5294, 5295).